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KIEV, PEREYASLAVL AND THE MIDDLE DNIEPER

Population and Towns of the Rus Heartland and the Steppe Frontier

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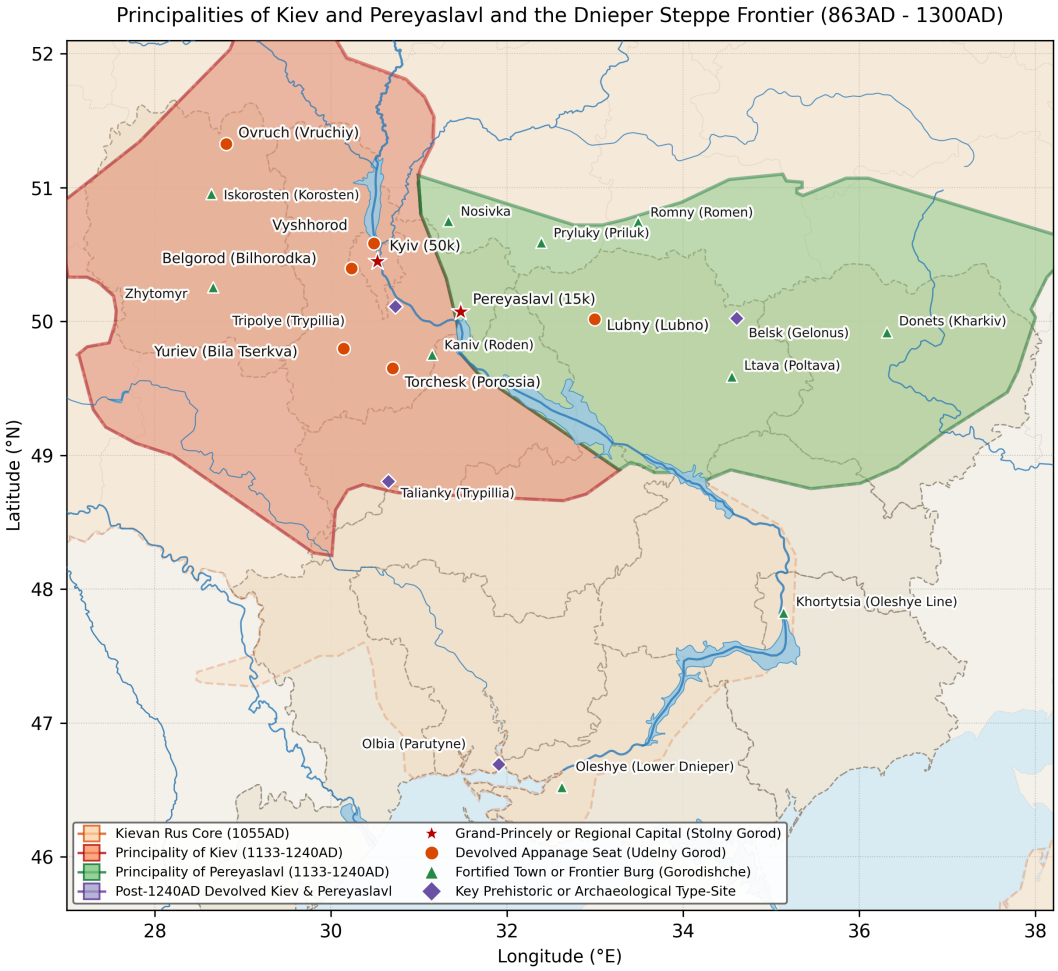


Figure 1. Approximate boundaries of the Principalities of Kiev and Pereyaslavl (~265k km^2), showing the metropolitan capital at Kiev (50k in 1200AD), the Left-Bank capital at Pereyaslavl-Russky (15k), the Porossia and Posullya rampart lines, the Drevlian and Bolokhov lands and the Lower Dnieper port of Oleshye.

Abstract

Where the forested rivers of Polesia meet the open black-earth steppe, the Middle Dnieper valley formed the historic heartland of Kievan Rus [1][2][3]. Across the pre-Mongol era, the region (~265k km^2) passed through four main waves of settlement growth. After the Eneolithic Trypillia culture built the great concentric towns of Talianky, Maidanetske and Nebelivka between the Ros and the Southern Bug around 3600BC (280k inhabitants) [4][5], Early Iron Age trade between Scythian forest-steppe hillforts such as Belsk (*Gelonus*, 4,02k ha) and the Greek port of Olbia raised the population to 540k around 500BC [6][7]. A third agricultural boom under the Roman-era Chernyakhov culture carried numbers to 720k around

300AD before the Hunnic invasions drove settlement back into the woods [8].

Under the Rurikid grand princes of the tenth to twelfth centuries, the construction of more than 950 km of timber-and-earth *Zmiyevy Valy* (Serpent's Walls) and 240 fortified towns (*gorodishcha*) along the Stuhna, Ros and Sula frontiers turned the Middle Dnieper into the most populous region of Eastern Europe, rising from 1,05M in 1000AD and 1,32M in 1100AD to 1,58M in 1200AD [1][2][9]. Kiev itself reached 50k inhabitants and Pereyaslavl 15k. This exposed forest-steppe frontier bore the full brunt of the Mongol conquest: Batu Khan's storming of Pereyaslavl in March 1239 and the sack of Kiev in December 1240 cut the regional population by more than half to 740k in 1250AD, and subsequent steppe raids held recovery to 820k by 1300AD [10][11].

Background

Geographically, the region covers the Middle Dnieper forest-steppe, the southern Pripyat, Teterev and Uzh woodlands of Polesia, the Upper Southern Bug basin and the Lower Dnieper steppe (~265k km^2 across modern central and southern Ukraine). Historically, this territory formed the core of unified Kievan Rus (863-1133AD) and later divided into two Rurikid principalities and their southern steppe marches (1133-1300AD).

The Principality of Kiev (1133-1322AD, ~95k km^2) occupied the Right-Bank Middle Dnieper plateau and the Irpin, Stuhna, Ros (*Porossia*), Teterev, Uzh and Sluch river basins (including the Drevlian Ovruch district), together with the farming country of the Bolokhov Land on the Upper Southern Bug and Horyn headwaters. Across the river, the Principality of Pereyaslavl (1133-1364AD, ~68k km^2) occupied the Left-Bank black-earth forest-steppe across the Trubezh, Supoy, Oster, Sula (*Posullya*), Khorol, Psyol, Vorskla and Upper Seversky Donets basins, serving as the eastern shield of Southern Rus against the Pechenegs, Torks and Polovtsians (Cumans). To the south, the Lower Dnieper and Pontic Steppe (~102k km^2) encompassed the Dnieper Rapids (*Porohy*), the Khortytsia-Protolche river crossing, the Kievan port of Oleshye (46,62°N, 32,62°E) at the Dnieper-Bug estuary and the winter pastures (*zimovniki*) of the Black Sea steppe between the Lower Southern Bug and the Azov coast.

Results

Prehistoric and Early Iron Age Settlement

Upper Palaeolithic, Neolithic and Trypillia Settlements

At the close of the Ice Age (10000BC), Epigravettian mammoth-hunters camped along the loess river terraces of the Dnieper, Ros, Supoy and Udai valleys. Excavations at Mezhirich (near Kaniv), Dobranichevka, Gontsy (on the Udai) and Kirillovskaya (in the Podol district of Kiev) reveal circular mammoth-bone dwellings housing winter bands of 30 - 60 hunters [4][12]. Across the Middle and Lower Dnieper basin, these hunting territories supported an estimated population of 14k at 10000BC. Following the transition to the Neolithic (6000BC), represented by the Bug-Dniester Culture on the Right Bank and the Dnieper-Donets Culture along the Dnieper Rapids (Nikolskoye, Vovnigi and Dereivka) and Left-Bank tributaries, fishing, foraging and early cattle-keeping communities raised the regional population to 42k.

During the Middle and Late Eneolithic (4200BC - 3200BC), the forest-steppe plateau between the Southern Bug, the Hnylyi Tikych and the Ros-Dnieper line became one of the most densely settled parts of Eneolithic Europe with the rise of the Cucuteni-Trypillia Culture (Stages BII and CI, peaking around 3600BC). Geomagnetic surveys and excavations directed by V. A. Kruts, M. Yu. Videiko and international teams have mapped a group of huge concentric settlements (*gigant-poseleniia*) on the Cherkasy-Uman

plateau that served as fortified farming towns [4][5][13]. Talianky (48,8°N, 30,53°E, Stage CI, c. 3700-3500BC) covered 450 ha with ~2,7k houses arranged in concentric ovals; assuming 75-80% of the houses were occupied at the same time by 5 - 7 persons per two-storey *ploshchadka* dwelling, Talianky held 15k inhabitants at 3600BC. Maidanetske (48,82°N, 30,69°E) enclosed 200 ha with ~2k - 2,3k houses arranged in four concentric rings and public assembly halls, supporting 10k inhabitants. Nebelivka (48,64°N, 30,56°E) covered 260 ha with a perimeter ditch, a two-storey temple and ~1,45k dwellings, supporting 9k inhabitants. Sister towns stood at Dobrovody (48,77°N, 30,38°E, 250 ha, 8,5k), Sushkivka (120 ha, 4,5k), Vesely Kut (150 ha, 5k), Kosenivka (120 ha, 4k), Chichirkozovka (100 ha, 3,5k) and Trypillia (Tripolye) on the Middle Dnieper bluffs (2,5k), alongside Late Trypillia CII Sofievka and Chapayevka cemeteries on the Kievan terraces [4].

Combining the Uman-Cherkasy large towns (~65k in settlements over 80 ha), the Right-Bank Middle Dnieper and Teterev farming villages (~145k) and the Early Usatovo and Lower Dnieper herding communities (~70k), the regional population reached an Eneolithic peak of 280k at 3600BC. With the drier climate of the late fourth millennium BC and the spread of Yamnaya (Pit-Grave), Catacomb and Middle Dnieper Corded Ware pastoralists by 2000BC, the great Trypillian towns gave way to mobile steppe herders and smaller riverside settlements (such as the copper-working and horse-breeding village at Dereivka and the fortified bluff at Pustynka), bringing the regional population to 195k in 2000BC.

Scythian Hillforts, Greek Poleis and the Chernyakhov Horizon

In the Early Iron Age (700BC - 300BC), trade in forest-steppe grain, hides, wax and slaves for Milesian Greek wine, olive oil and metalwork through the Lower Bug-Dnieper estuary turned the Middle Dnieper into the farming heartland of Herodotus's Scythian ploughmen (*Skolotoi*), Geloni and Budini [6][14]. Across the Left-Bank Vorskla watershed, the Belsk Hillfort (identified by B. A. Shramko with Herodotus's Gelonus, 50,09°N, 34,63°E) enclosed 4,02k ha within 33,8 km of earthen ramparts uniting the Western (72 ha), Eastern (65,2 ha) and Kuzeminsk (15,4 ha) forts, housing bronze-casting, iron-smelting and Greek pottery workshops with an estimated population of 12k [6]. On the Right Bank and Left-Bank Sula-Trubezh rivers stood a ring of large hillforts: Trakhtemyriv on the Kaniv Dnieper bend (500 ha, 4,5k), Khotiv on the southern outskirts of Kiev (31 ha, 2,5k), Motronyn in the Cherkasy Cold Ravine (200 ha, 4k), Pastyrskoye (25 ha, 2,5k), Nemirov on the Southern Bug (110 ha, 4k), Basovka on the Upper Sula (170 ha, 3k) and Knyazha Hora (1,8k) [14]. At the same time, along the Lower Bug-Dnieper coast, Milesian colonists in the mid-seventh to early sixth centuries BC founded Pontic Olbia (Parutino, 46,69°N, 31,9°E, a 50 ha stone-walled upper and lower town surrounded by over 100 rural villages along the Berezan-Bug-Dnieper estuary), which reached 15k inhabitants in the town itself at 500BC, accompanied by the island trading station of Berezan (Borysthenes, 2,5k), Nikonion (2k) and Glubokaya Pristan (1,2k) [7].

Together with open farming villages (*selishcha*) across the Kiev-Cherkasy plateau and royal Scythian winter camps in the Lower Dnieper (forerunners of Kamenskoye Gorodishche, 1,2k ha, which peaked in the fourth century BC at 6k), the Middle and Lower Dnieper supported 540k inhabitants at 500BC.

Following the Sarmatian raids of the third and second centuries BC, the Zarubintsy Culture (200BC - 100AD, named after the cemetery near Trakhtemyriv and Pereyaslavl, with key settlements at Pilipenkova Hora at Kaniv, Chodosovka, the Liutezh ironworks north of Kiev and Subotiv) maintained a riverine population of 410k at 0AD alongside the late Scythian stone forts of the Lower Dnieper (Zolota Balka, Gavrilovka, Lyubimovka and Znamenskoye) [15]. During the Late Roman period (250AD - 375AD), the Chernyakhov Culture introduced heavy iron ploughshares, rotary querns and wheel-thrown grey pottery across more than 1,2k unfortified valley villages in the Kiev, Pereyaslavl and Cherkasy forest-steppe (Obukhiv, Zhukivtsi, Chernyakhiv, Bilozerka, Kompaniiets and Velikaya Bugaevka). B. V. Magomedov estimates an average village size of 15 - 25 homesteads (90 - 150 inhabitants) spaced every 2 - 4 km along minor streams, bringing the regional population to a pre-medieval peak of 720k at 300AD [8]. The Hunnic

invasion of 375AD and the collapse of Roman Black Sea trade caused a sharp drop; by 500AD, the Early Slavic Prague-Korchak Culture (in the Right-Bank Polesian and Teterev woods) and the Penkovka Culture (associated with the *Antae* along the Ros, Tyasmin, Sula and Psyol), together with the early hilltop settlement on Starokyivska Hora (Kiev, 1,5k at 500AD) and the smithing centre of Pastyrskoye (2,5k in the sixth and seventh centuries AD), sustained 430k inhabitants [16][17].

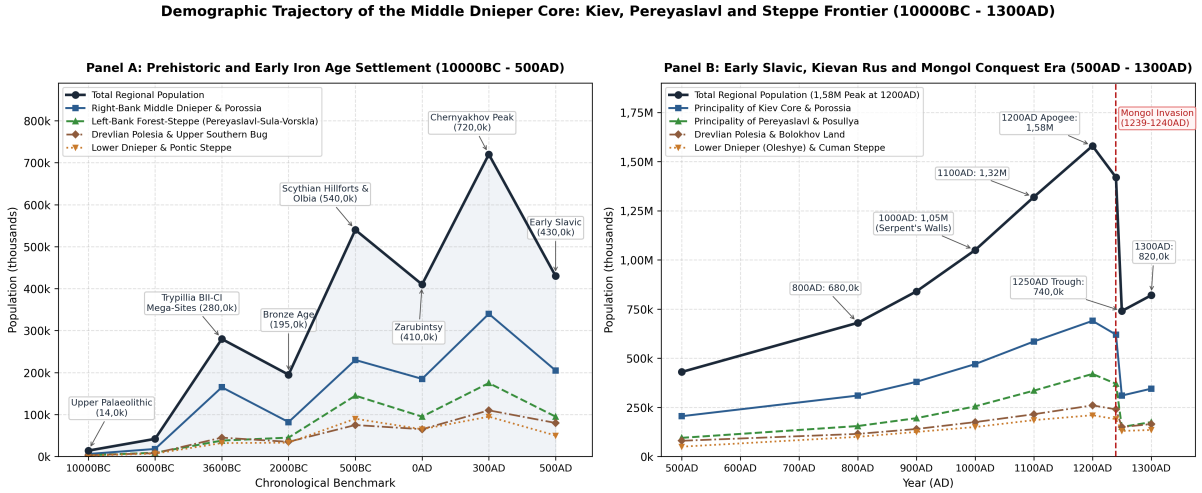


Figure 2. Shaded bands indicate uncertainty ranges around the central estimate for the Middle and Lower Dnieper (265k km^2). Pre-Slavic settlement peaked during the Trypillia BII-CI concentric towns (280k at 3600BC), the Scythian-Olbian trade era (540k at 500BC) and the Roman-era Chernyakhov culture (720k at 300AD); medieval growth reached 1,58M in 1200AD before Batu Khan's campaigns of 1239-1240AD reduced the population by more than half to 740k in 1250AD.

Date	Right-Bank Kiev & Porossia	Drevlian & Bolokhov Land	Left-Bank Pereyaslavl & Sula	Lower Dnieper & Steppe	Regional Total
10000BC	6k	3k	3k	2k	14k
6000BC	18k	8k	9k	7k	42k
3600BC	165k	45k	38k	32k	280k
2000BC	82k	35k	45k	33k	195k
500BC	230k	75k	145k	90k	540k
0AD	185k	65k	95k	65k	410k
300AD	340k	110k	175k	95k	720k
500AD	205k	80k	95k	50k	430k

Table 1. Sub-regional population estimates (in thousands) across the four geographic zones of the Middle and Lower Dnieper from the Late Palaeolithic mammoth-hunter camps (10000BC) to the early Slavic Prague-Korchak and Penkovka horizon (500AD).

Early Slavic and Unified Kievan Rus

Luka-Raikovetska, Volyntsevo and the Rise of Kiev

During the eighth and ninth centuries AD, Eastern Slavic farming villages multiplied on both banks of the Middle Dnieper. On the Right Bank, the Luka-Raikovetska Culture built fortified tribal hillforts (*grady*) and open villages across the Polanian (*Poliane*), Drevlian (*Drevliane*) and Ulichian (*Ulichi*) lands, including Starokyivska Hora, Zamkova Hora and the earliest tree-ring-dated log houses in Podol (887AD layer at Kiev, 5k in 800AD, rising to 12k in 900AD) [1][18], Iskorosten (Korosten) on the Uzh (1,8k in 800AD), Malyn, Raiky and Monastyrek on the Kaniv Dnieper bend (1,2k). On the Left Bank, the Volyntsevo-Romny Culture of the Severians (*Severiane*) occupied the Desna-Seim-Sula-Psyol forest-steppe (Opishnia, Bititsa at 2k, Romny or *Romen* at 1,2k and Donets at 1k), paying tribute first to the Khazar Khaganate before Oleg of Novgorod united the Dnieper waterway in 882AD [19][20]. The regional population recovered to 680k in 800AD and reached 840k in 900AD as Scandinavian-Slavic *druzhina* retinues organised the *polyudie* winter tribute round linking the Drevlian, Polanian and Severian rivers to the Byzantine trade treaties of 907AD and 911AD.

Vladimir the Great, Yaroslav the Wise and the Serpent's Walls

The conversion of Rus under Vladimir Svyatoslavich (980-1015AD) and the building programme of Yaroslav the Wise (1019-1054AD) reshaped the Middle Dnieper through frontier fortifications and military settlement. As recorded in the *Primary Chronicle* (*Povest' vremennykh let*, under the years 988AD and 991AD), Vladimir declared that it was not good that there were so few towns around Kiev, and built defensive lines along the Stuhna, Irpin, Trubezh, Oster, Ros and Sula rivers, garrisoning them with warriors brought down from the Slovenes of Novgorod, the Krivichi, the Chud and the Vyatichi [21][22].

Archaeological excavations by M. P. Kuchera across the Serpent's Walls (*Zmiyevy Valy*) show that between the late tenth and mid-eleventh centuries, the Kievan state built over 950 km of continuous timber-laced earthen ramparts (10 - 12 m high, fronted by 5 - 6 m ditches) in six lines: the Vita-Bobrytsia, Stuhna, Irpin-Teterev, Ros (Porossia), Pereyaslavl-Trubezh and Sula (Posullya) ramparts [9]. Sheltered behind these walls, the number of rural villages (*selishcha*) tripled between 950AD and 1100AD. By 1000AD (1,05M regional total), Kiev had grown from Vladimir's 10 ha *Detinets* (around the Church of the Tithes, consecrated 996AD) and the artisans' quarter of *Podol* to 28k inhabitants, ringed by the fortified royal residences and garrison towns of Belgorod (110 ha enclosure founded 991AD, 6k), Vyshhorod (70 ha, 5k), Vasiliev (Vasylkiv, 2,5k), Trepol (Trypillia, 2k), Vitichev (1,8k) and the Left-Bank capital of Pereyaslavl-Russky (80 ha fortified town built 992AD, 8k) [1][23]. By 1100AD (1,32M regional total), under Yaroslav the Wise (who built the 70 ha *City of Yaroslav* enclosed by 3,5 km of 16 m-high ramparts, the Golden Gate and St Sophia Cathedral in 1037AD, and founded the Porossia frontier line along the Ros in 1032AD around Yuriev or Bila Tserkva, Torchesk and Korsun) and Vladimir Monomakh (who led the campaigns against the Cumans from Pereyaslavl after the Congresses of Lyubech in 1097AD and Dolobsk in 1103AD), Kiev reached 40k inhabitants and Pereyaslavl-Russky reached 12k [3][24].

Devolved Principalities

Following the death of Mstislav the Great (1132AD), the unified realm divided into separate dynastic lands (1133AD - 1240AD). In 1200AD, at the population peak of pre-Mongol Southern Rus, the Middle and Lower Dnieper region reached 1,58M inhabitants across 240 excavated fortified towns (*gorodishcha*) catalogued by A. V. Kuza [2][23] and O. P. Motsia [3] and more than 2,8k open rural villages (*selishcha*).

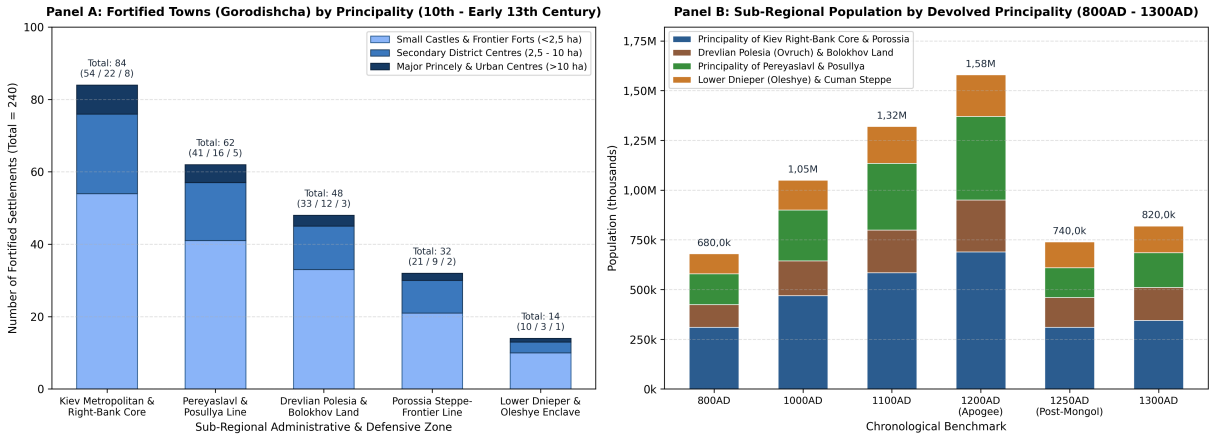


Figure 3. Left: distribution of the 240 excavated fortified settlements (gorodishcha) catalogued by Kuza and Motsia across the five Middle and Lower Dnieper zones, distinguishing major towns (>10 ha), medium fortresses (2,5 - 10 ha) and small border forts (<2,5 ha). Right: sub-regional population shares from the Luka-Raikovetska and Volyntsevo tribal era (800AD) through the 1200AD peak to the post-Mongol contraction.

The City of Kiev

Excavations by P. P. Tolochko [1][12][24], M. A. Sahaydak [18] and H. Yu. Ivakin [10] show that late twelfth- and early thirteenth-century Kiev covered 360 - 400 ha (380 ha in our estimates). The Upper City (*Gora*, ~98 ha) consisted of *Vladimir's City* (10-12 ha, with princely *dvor* halls, stone churches and boyar compounds), *Yaroslav's City* (70 ha, around the metropolitan cathedral of St Sophia, St Irene, St George and dense housing quarters), *Izyaslav-Svyatopolk's City* (8 ha, around St Michael's Golden-Domed Monastery) and *Kopyrev Konets* (8 ha on the north-western spur). Below the bluffs, the Lower City (*Podol*, 180 - 200 ha) has been uncovered across *Kontraktova Ploshcha*, *Zhytniy Rynok*, *Verkhniy Val* and *Nyzhniy Val*, revealing fenced urban homesteads (*usadby*) of 300 - 600 m^2 containing 2 - 3 log dwellings, craft workshops (glass-making, bone-carving, enamelling, shoemaking and smithing) and merchant warehouses (*pirohy*). Allowing 3,5k - 4k occupied homesteads across *Podol* at 6 persons per household gives 22k - 25k residents in the Lower City alone [18]. Around the walled core lay suburban monasteries and princely estates (~80 - 100 ha), including the *Kievo-Pecherskaya Lavra* (Caves Monastery), the princely seat of *Berestovo*, *Klov*, *Dorogozhichi*, *Vydubychi*, *Kyrylivska*, *Kitaevo* (a 35 ha fort on the southern Dnieper hills) and *Gnilets*.

Combining the Upper City boyar and clerical households (~15k), the *Podol* artisan and merchant homesteads (~24k) and *Kopyrev Konets* and the suburbs (~9k - 11k), Tolochko's estimate of 50k inhabitants at 1200AD (~131,6 persons/ha across 380 ha) places Kiev among the largest cities of twelfth-century Europe alongside Constantinople, Paris and London [24][25].

The Right-Bank Kievan Ring and Stuhna-Dnieper Corridor

Protecting the southern and western approaches to Kiev stood a ring of princely seats, river ports and forts (84 fortified *gorodishcha* in the Kiev Core zone: 8 major towns over 10 ha, 22 medium towns of 2,5-10 ha and 54 small castles under 2,5 ha). Belgorod (Bilhorodka) on the Irpin (50,39°N, 30,23°E, 110 ha *detinets* and *okolny gorod*, bishopric and second capital of Rurik Rostislavich) held 9k inhabitants at 1200AD. Vyshhorod (50,58°N, 30,49°E, 70 ha cliffside citadel and potters' *posad*, housing the shrine of SS Boris and Gleb) had 7k, while Kaniv (49,75°N, 31,47°E, 24 ha, Dnieper crossing and muster point against the Cumans,

with the stone Church of St George built in 1144AD) reached 4k. Other fortified towns included Vasiliev (Vasylkiv) on the Stuhna (20 ha, 3,5k), Trepol (Trypillia, 18 ha Dnieper cliff fortress guarding the Stuhna mouth, 3k), Roden on *Knyazha Hora* (16 ha craft town near Kaniv, 2,5k), Vitichev (14 ha signal fortress on the Dnieper, 2,2k), Zarub (1,5k), Chuchin (12 ha, 1,8k), Khalep (1,4k), Svyatoslavl, Ivan-Gorod (1,5k), Tumasch (1,4k), Kitaigorod (1,5k) and Zvenyhorod-Kievan (1,6k) [2][3].

The Porossia Steppe-Frontier Line

Along the Ros River valley (80 km south of Kiev), the Kievan grand princes settled Turkic border horsemen, the Chorni Klobuky (“Black Hoods”, uniting the Torks, Berendei, Kovui, Turpei and remaining Pechenegs), in 32 fortified towns (2 major, 9 medium and 21 small) [3][22]. Torchesk (49,68°N, 30,63°E, identified with the 45 ha *Olshanitsa* or *Sharki* hillfort on the Gorokhovatka-Ros) served as the capital of the Black Hoods and appanage seat of Mstislav the Bold (1193-1207AD), holding 6k inhabitants at 1200AD. Yuriev (Bila Tserkva, 49,79°N, 30,11°E, 28 ha *Zamkova Hora* and *posad*, seat of the Bishop of Yuriev) housed 4,5k, flanked by Korsun on the Ros (49,42°N, 31,26°E, 16 ha, 2,8k), Boguslavl (49,55°N, 30,87°E, 15 ha, 2,5k), Volodarev (Volodarka, 1,8k), Rostovets (1,8k), Dveren (1,5k), Polksten (1,5k), Bakhmutov (1,2k), Sakkov (1,1k) and Borovyi (1k). Together, the Right-Bank Kiev Core and Porossia frontier supported 690k inhabitants at 1200AD.

Drevlian Polesia and the Bolokhov Land

North-west and south-west of the Kievan core lay two provinces (~260k inhabitants at 1200AD) containing 48 fortified *gorodishcha* (3 major over 10 ha, 12 medium of 2,5-10 ha and 33 small under 2,5 ha). In Drevlian Polesia, settlement centred on Ovruch (Vruchiy, 51,32°N, 28,8°E, 25 ha, with the brick Church of St Basil built by Rurik Rostislavich c. 1190AD, 4,5k at 1200AD) and Iskorosten (Korosten, 50,95°N, 28,64°E, a four-citadel town on the Uzh, 22 ha, 4k). As shown by A. P. Tomashevsky [26], the Ovruch ridge was the sole European source of pink pyrophyllite slate, ringed by more than 40 quarry and workshop villages (*Nagorany*, *Kamenka* and *Korenevka*) carving slate spindle-whorls (*priaslitsa*) and sarcophagi exported across Rus, Poland and Volga Bulgaria. Smaller Drevlian towns included Vozvyagl (Zviahel) on the Sluch (16 ha, 2,8k), Malyn (1,8k), Kotelnich (2k), Zhytomyr (1,8k), Horodsk (*Gorodesk*, 2k), Michesk (Radomyshl, 1,6k), Ushchesk (Ushomyr, 1,5k) and Chornobyl (1,5k at the Pripyat-Uzh confluence). To the south-west, the Bolokhov Land (*Bolokhovskaia Zemlia*, 1150-1257AD) occupied the fertile forest-steppe plateau of the Upper Southern Bug, Sluch and Horyn between the principalities of Kiev, Galicia-Volhynia and the Cuman steppe; combining the Right-Bank core (690k) with the Drevlian and Bolokhov lands (260k) gives 950k across the Principality of Kiev at 1200AD. Ruled by local princes (*knyaztsi bolokhovskie*), its fortified towns, including Bolokhov (Lyubar, 15 ha, 2,5k), Gubin (12 ha, 2k), Kudin (1,8k), Kobud (1,6k), Derevich (1,6k), Bozhsky (*Bozhesk*, 2k), Mezhibozhye (Medzhybizh, 2,2k) and Dyadkov (1,4k), grew wheat and millet for the surrounding lands [3][27].

Principality of Pereyaslavl

East of the Dnieper, the Principality of Pereyaslavl (~420k inhabitants at 1200AD) served from the eleventh to early thirteenth centuries as the ancestral domain of the Monomakhovichi and the main military shield of Southern Rus [19][28]. Surveys by Yu. Yu. Morgunov and O. V. Sukhobokov record 62 fortified *gorodishcha* across the Pereyaslavl land (5 major over 10 ha, 16 medium of 2,5-10 ha and 41 small under 2,5 ha). Built at the junction of the Alta and Trubezh rivers, the capital of Pereyaslavl-Russky (Pereiaslav, 50,07°N, 31,46°E) had a 10 ha *Detinets* (containing Bishop Ephraim’s stone Cathedral of St Michael built in 1089AD, stone episcopal gates, a bathhouse and a princely palace) and a 70 ha walled *Okolny Gorod* plus open riverside *posady* (80 ha walled area, ~100 ha in all), holding 15k inhabitants at 1200AD [1][2].

Stretching 180 km from the Sula mouth on the Dnieper to the Upper Sula border with the Kursk-Putivl

Severians, the Posullya frontier line consisted of 18 inter-visible timber-and-earth forts (*gorodishcha*) anchored at the river mouth by the fortified harbour of Voin (Voinska Hreblia, 18 ha with a dredged inner ship-basin, 2,5k at 1200AD), followed upstream by Zheldi (1,4k), Rimiv (14 ha, besieged by Konchak in 1185AD, 2,2k), Lukoml (13 ha, 2k), Lubno (Lubny, 16 ha, founded by Monomakh, 3k), Khorol (2k), Sneporod (12 ha, 1,8k), Ksniatin (13 ha, 2k), Sinets (1,5k), Romen (Romny, 15 ha, 2,5k) and Glinsk (1,5k) [28]. Behind the Sula line in the interior Trubezh, Supoy and Oster valleys lay Priluk (Pryluky) on the Udai (15 ha, 2,8k), Nosivka (2,2k), Oster (Gorodets-Ostersky, 14 ha, founded by Monomakh in 1098AD with St Michael's *Yurievska Bozhnytsia*, 2,5k), Baruch (1,8k), Bronnitsa (1,6k), Pesochen (1,5k) and Zholnin (1,4k), while the south-eastern forest-steppe along the Psyol, Vorskla and Seversky Donets held Ltava (Poltava, 49,58°N, 34,56°E, 14 ha on *Ivanova Hora*, mentioned 1174AD, 2,2k), Goltav (1,5k), Nizhegolsk (1,4k) and the easternmost frontier fortress of Donets (49,85°N, 36,21°E, the Khorosheve hillfort south of modern Kharkiv, where Igor Svyatoslavich escaped from Polovtsian captivity in 1185AD, 2k) [19][29].

Lower Dnieper Ports and the Steppe Frontier

South of the Porossia and Posullya ramparts lay the Pontic Steppe (*Pole Polovetskoe*, ~210k inhabitants at 1200AD), where 14 fortified river-island and delta trading stations shared the country with the seasonal camps (*vezhi* and winter *zimovniki*) of the Dnieper Polovtsian hordes (the Burchevichi and Lukomorie Cumans led by Khans Kobyak, Bonyak and Kotyan) [3]. At Oleshye (46,62°N, 32,62°E, on Veliky Potemkin Island opposite modern Kherson and Oleshky), the sea gate of Kievan Rus covered a 20 ha waterlogged port yielding thousands of Byzantine amphorae, lead seals of Kievan princes and Greek officials, glass bracelets and ship fittings; here Kievan war fleets met Byzantine merchants and newly consecrated Metropolitans, supporting 3,5k permanent inhabitants at 1200AD. Upstream at Khortytsia (Protolche, 47,82°N, 35,07°E), the muster station below the Dnieper Rapids (*Khortytskyi Ostrov* and *Protolche* ford), where the Rus princes gathered before the 1103AD Suten and 1223AD Kalka campaigns, held 1,8k inhabitants at 1200AD, alongside Berestovo (Beloberezhye, 1,2k) on the Kinburn spit and Cuman winter towns along the Samara, Orel and Molochnaya rivers (210k in all across the Lower Dnieper and steppe at 1200AD).

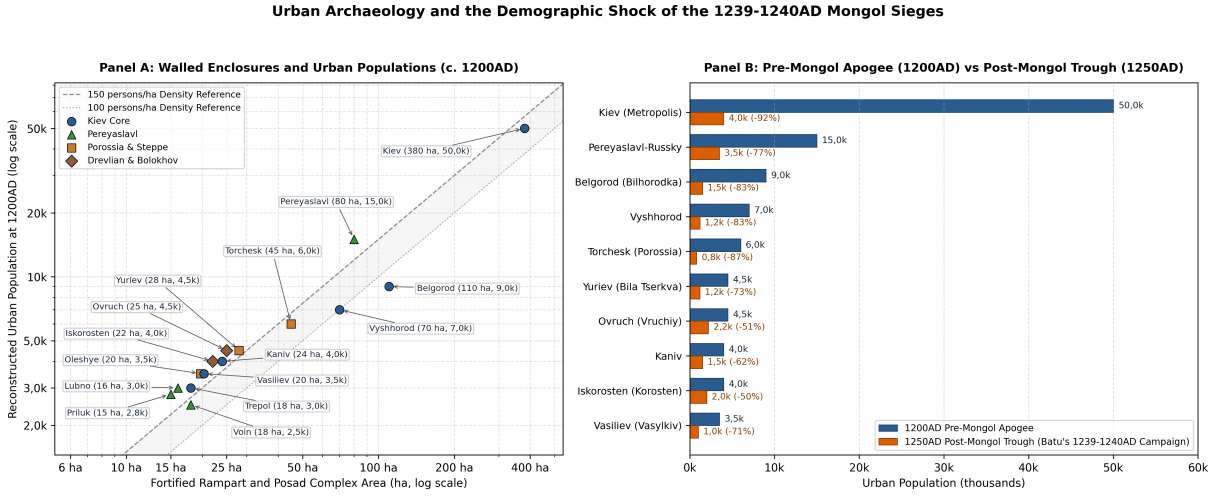


Figure 4. Left: walled rampart and suburban posad areas plotted against urban population in 1200AD, led by the 380 ha metropolis of Kiev (50k) and the 80 ha Left-Bank capital of Pereyaslavl-Russky (15k). Right: the demographic collapse of the exposed Middle Dnieper capitals and frontier towns between 1200AD and 1250AD compared with the partial survival of wooded Drevlian Polesia (Ovruch and Iskorosten).

Mongol Invasion and Post-Siege Recovery

Batu Khan's Campaigns of 1239-1240AD

After the raid of Jebe and Subutai and the defeat of the Kievan, Galician, Chernigov and Smolensk princes at the Battle of the Kalka River (31 May 1223AD), where Grand Prince Mstislav Romanovich of Kiev and 10k Kievan warriors fell, the Middle Dnieper principalities entered 1240AD with a slightly reduced population of 1,42M. The Mongol conquest of Southern Rus followed in two campaigns [1][10][29].

In March 1239AD, Mongol armies under Möngke and Batu broke through the Posullya ramparts, burning Rimiv, Voin, Lubno, Lukoml and Priluk, and stormed Pereyaslavl-Russky on 3 March 1239AD. As recorded in the *Laurentian* and *Hypatian Chronicles*, St Michael's Cathedral was wrecked, Bishop Simeon was killed and the townspeople were slaughtered or led away into captivity. Excavations across the Pereyaslavl *Detinets* and the *Posullya* hillforts uncover a continuous 1239AD burn layer filled with Mongol armour-piercing arrowheads, unburied skeletons in burned cellars and silver hoards (*klady*) whose owners never returned [28]. After Möngke reconnoitred Kiev from *Pesochen (Gorodok)* on the Left Bank in late 1239AD, Batu Khan's main host crossed the Dnieper in autumn 1240AD, overrunning the Porossia border towns (Torchesk, Yuriev, Korsun, Vitichev, Vasiliev, Belgorod and Vyshhorod). Defended by Daniel of Galicia's commander Dmitr, Kiev was ringed by Mongol catapults (*poroki*), which breached the Lyadskie Gates of Yaroslav's City before a final stand around the Church of the Tithes, whose vaults collapsed under the weight of defenders and refugees on 6 December 1240AD.

Mass graves excavated by M. K. Karger, P. P. Tolochko and H. Yu. Ivakin in the Kiev Upper City (over 2k skeletons in pits near the Church of the Tithes and St Sophia, alongside burned workshops with slain craftsmen inside) show the scale of the slaughter [1][10]. When the Papal envoy Giovanni da Pian del Carpine (John of Plano Carpini) rode through the Kievan land in February 1246AD, he wrote:

> *"They attacked Rus, where they made great havoc, destroying cities and fortresses and slaughtering men; and they laid siege to Kiev, the capital of Rus; after they had besieged the city for a long time, they took it and put the inhabitants to death. When we were journeying through that land we came across countless skulls and bones of dead men lying about on the ground. Kiev had been a very large and densely populated town, but now it has been reduced almost to nothing, for there are at the present time scarce two hundred houses there and the inhabitants are kept in utter slavery."* [11]

Allowing for Carpini's ~200 taxable house compounds (*dvory*) on *Podol* plus surviving clerical and suburban households around the Caves Monastery and *Kopyrev Konets*, H. Yu. Ivakin puts Kiev's population at 3,5k - 4,5k (4k in our tables) in 1250AD, down 92% from its 1200AD peak of 50k [10].

The Bolokhov Land and Late Thirteenth-Century Recovery

Losses during the 1239-1240AD invasion fell unevenly across the Middle Dnieper. The heaviest devastation hit the open forest-steppe (-64% in Pereyaslavl and Posullya; -55% in Kiev Core and Porossia): along the Sula and Ros frontier lines, nearly four-fifths of the fortified *gorodishcha* (Voin, Rimiv, Sneporod, Torchesk, Roden and Vitichev) were abandoned for good after 1239-1240AD [2][28]. By 1250AD, the total regional population had fallen to 740k: 310k in the Right-Bank Kiev Core and Porossia, 150k in the Drevlian and Bolokhov lands, 150k in the Principality of Pereyaslavl and 130k in the Lower Dnieper and steppe.

Forested Drevlian Polesia and the Bolokhov Land fared better. The marshy woods of the Teterev, Uzh and Pripyat valleys (Ovruch, Iskorosten, Chornobyl and Zhytomyr) sheltered countryfolk fleeing the Dnieper valley. Meanwhile, the towns of the Bolokhov Land (Bolokhov, Gubin, Kudin, Kobud and Bozhsky) escaped burning in 1240AD by submitting directly to the Mongols as tribute-paying ploughmen

(*Tataropoli*), growing wheat and millet for Batu's armies until King Daniel Romanovich of Galicia-Volhynia marched against them in 1241AD and 1255-1257AD and levelled their ramparts along with Vozvyagl [3][27].

After the Golden Horde censuses (*chislo*) of 1257-1259AD and 1273AD under Mengu-Timur, the Principalities of Kiev and Pereyaslavl were governed by local Rurikid and Putivl-Olgovichi princes under the watch of Golden Horde tax-collectors (*baskaki*). Although Metropolitan Maximus moved the seat of the Metropolitanate of All Rus from Kiev to Vladimir-on-the-Klyazma in 1299AD, Ivakin's excavations show a steady late thirteenth-century revival of crafts and river trade in Kievan *Podol*, *Pechersk* and the castle on *Zamkova Hora* (8,5k inhabitants in 1300AD), as well as in Pereyaslavl (4,5k), Ovruch (3k), Iskorosten (2,6k), Kaniv (2,2k) and Yuriev (2k), lifting the regional population back to 820k by 1300AD on the eve of the Lithuanian conquest under Gediminas in the 1320s [10].

Date	Kiev Core & Porossia	Drevlian & Bolokhov Land	Pereyaslavl & Posullya	Lower Dnieper & Steppe	Regional Total
500AD	205k	80k	95k	50k	430k
800AD	310k	115k	155k	100k	680k
900AD	380k	140k	195k	125k	840k
1000AD	470k	175k	255k	150k	1,05M
1100AD	585k	215k	335k	185k	1,32M
1200AD	690k	260k	420k	210k	1,58M
1240AD	620k	240k	370k	190k	1,42M
1250AD	310k	150k	150k	130k	740k
1300AD	345k	165k	175k	135k	820k

Table 2. Sub-regional population estimates (in thousands) across the Principalities of Kiev and Pereyaslavl, the Drevlian-Bolokhov uplands and the Lower Dnieper frontier from the sixth-century Antae and Polanian settlements (500AD) to the Golden Horde appanage era (1300AD).

Comparison with Other Rus Principalities

Table 3 compares the population of the Middle Dnieper (Kiev and Pereyaslavl) with the northern and eastern Rus principalities between 1000AD and 1300AD [21][23][30].

Principality / Region	1000AD Pop.	1100AD Pop.	1200AD Peak	1250AD Low	1300AD Pop.
Principality of Kiev (incl. Drevlian & Bolokhov)	645k	800k	950k	460k	510k
Principality of Pereyaslavl (incl. Posullya)	255k	335k	420k	150k	175k
Lower Dnieper (Oleshye) & Pontic Steppe Frontier	150k	185k	210k	130k	135k
Kiev and Pereyaslavl Subtotal	1,05M	1,32M	1,58M	740k	820k
cf. Vladimir-Suzdal/Rostov (European Russia)	340k	510k	780k	520k	640k
cf. Novgorod & Pskov Republics (European Russia)	290k	380k	480k	460k	510k

cf. Principality of Smolensk (European Russia)	210k	275k	340k	290k	310k
cf. Kursk-Rylsk & Bryansk-Upper Oka (European Russia)	280k	390k	500k	270k	310k
cf. Total European Russia Benchmark	2,15M	2,9M	3,85M	2,8M	3,1M

Table 3. Population of the Kievan and Pereyaslavl principalities compared with Vladimir-Suzdal, Novgorod-Pskov, Smolensk and the northern Chernigov-Seversk appanages, 1000AD - 1300AD. Whereas European Russia lost 27,3% of its population between 1200AD and 1250AD, the exposed Middle Dnieper heartland fell by 53,2%.

Conclusion

Until the eve of the Mongol invasion, the Middle Dnieper remained the most populous and urbanised heartland of the Rus world (1,58M inhabitants in 1200AD), containing two of its five largest cities (Kiev at 50k and Pereyaslavl at 15k) and 240 fortified towns. Yet the same open forest-steppe terrain that made the black-earth river valleys so productive also left them exposed to Batu Khan's siege armies in 1239-1240 and to decades of Golden Horde border raids thereafter. Whereas the population of European Russia fell by 27,3% between 1200AD and 1250AD (from 3,85M to 2,8M, cushioned by the uninvaded forests of Novgorod, Pskov and Smolensk) and recovered to four-fifths of its pre-Mongol level by 1300AD (3,1M), the Middle Dnieper lost more than half its people (falling 53,2% from 1,58M to 740k in 1250AD) and by 1300AD had regained little more than half of its old numbers (820k). The centre of gravity of Eastern Slavic population had shifted permanently to the northern forests and the Carpathian west.

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